

1.

$$m(a+bX) = \frac{1}{N} \sum_{i=1}^N (a+bx_i)$$

$$= \frac{1}{N} \left(\sum_{i=1}^N a + \sum_{i=1}^N bx_i \right)$$

$$= \frac{1}{N} (N \cdot a) + \frac{b}{N} \sum_{i=1}^N x_i$$

$$= a + b \cdot m(X)$$

2. $\text{cov}(X, a+bY) = \frac{1}{N} \sum_{i=1}^N (x_i - m(X))(a+by_i) -$
 $(a+b \cdot m(Y))$

$$= \frac{1}{N} \sum_{i=1}^N (x_i - m(X)) \cdot b(y_i - m(Y))$$

$$= bX \left(\frac{1}{N} \sum_{i=1}^N (x_i - m(X))(y_i - m(Y)) \right)$$

$$\begin{aligned}
 3. \quad \text{cov}(a+bx, a+bx) &= \frac{1}{N} \sum_{i=1}^N [b(x_i - m(x))][b(x_i - m(x))] \\
 &= \frac{1}{N} \sum_{i=1}^N b^2 (x_i - m(x))^2 \\
 &= b^2 \text{cov}(X, X)
 \end{aligned}$$

$$\begin{aligned}
 \text{cov}(X, X) &= s^2 \\
 &= \frac{1}{N} \sum_{i=1}^N (x_i - m(x))(x_i - m(x)) \\
 &= \frac{1}{N} \sum_{i=1}^N (x_i - m(x))^2 = s^2
 \end{aligned}$$

4. Not always true ; If $g(x) = \text{convex}$
 $m(g(x)) \geq g(m(x))$

If $g(x) = \text{concave}$
 $m(g(x)) \leq g(m(x))$